1.) (This is HW3 # 1e.)

Fix $a, b \neq 0$. Then

$$\begin{aligned}(a^{-1}\cdot b^{-1})\cdot(ab) &= (a^{-1}\cdot b^{-1})\cdot(ba)\\ &= (a^{-1}\cdot(b^{-1}\cdot b))\cdot a\\ &= (a^{-1}\cdot 1)\cdot a\\ &= a^{-1}\cdot a\\ &= \underline{1}\end{aligned}$$

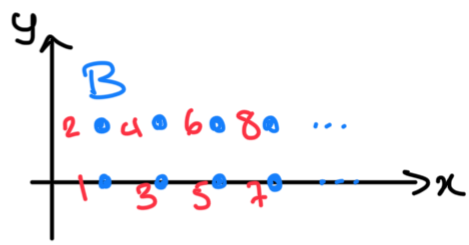
So $a^{-1}\cdot b^{-1}$ is the multiplicative inverse $(ab)^{-1}$ of ab. $\square$

2.) (This is a special case of HW2 # 1.)

(a.) $\exists\ f: A \to \mathbb{N}$ bijective

(b.) Define $f: B \to \mathbb{N}$ by

$$f(x,y) = \begin{cases} 2x-1 & \text{if } y=0 \\ 2x & \text{if } y=1 \end{cases}$$



Want: f bijective. Consider $g: \mathbb{N} \to B$,

$$g(n) = \begin{cases} \left(\frac{n+1}{2}, 0\right) & \text{if } n \text{ odd} \\ \left(\frac{n}{2}, 1\right) & \text{if } n \text{ even} \end{cases}$$

Then

$$(f\circ g)(n) = \begin{cases} f\left(\frac{n+1}{2}, 0\right) = 2\left(\frac{n+1}{2}\right)-1 = n & \text{if } n \text{ odd} \\ f\left(\frac{n}{2}, 1\right) = 2\left(\frac{n}{2}\right) = n & \text{if } n \text{ even} \end{cases}$$

$$(g\circ f)(x,y) = \begin{cases} g(2x-1) = \left(\frac{(2x-1)+1}{2}, 0\right) = (x,0) & \text{if } y=0 \end{cases}$$

$$\lfloor g(2x) = \left(\tfrac{2x}{2}, 1\right) = (x, 1) \qquad \text{if } y=1$$

for all $n \in \mathbb{N}$ and $(x,y) \in B$. So f is bijective by HW. $\square$

3.) (This is HW3 #5.)

As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, $\sup A$, $\sup B \in \mathbb{R}$ exist.

① $\sup A + \sup B$ is an upper bound for S:

As $\sup A$, $\sup B$ are upper bounds for A, B,

$$a \leq \sup A \quad \forall a \in A, \qquad b \leq \sup B \quad \forall b \in B$$

$$\Rightarrow \quad a+b \leq \sup A + \sup B \quad \forall a \in A, \ \forall b \in B$$

② L upper bound for $S \Rightarrow L \geq \sup A + \sup B$:

Suppose not. Then L is an upper bound for S, but $L < \sup A + \sup B$. Set

$$\varepsilon = \tfrac{1}{2}(\sup A + \sup B - L) > 0.$$

As $\sup A - \varepsilon < \sup A$, then $\exists\, a_0 \in A$ s.t.

$$a_0 > \sup A - \varepsilon.$$

As $\sup B - \varepsilon < \sup B$, then $\exists\, b_0 \in B$ s.t.

$$b_0 > \sup B - \varepsilon.$$

Together,

$$a_0 + b_0 > \sup A + \sup B - 2\varepsilon = L.$$

This contradicts that L is an upper bound for S.

Collecting ① and ②, we conclude $\sup S$

$= \sup A + \sup B.$ $\square$

4.) (This is HW4 #1.)

(a.) <u>Prove</u>: Fix $\varepsilon > 0$. As $\lim_{n\to\infty} a_n = a$, $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \varepsilon \quad \forall n \geq N$$

Then for $n \geq N$ we have

$$\big||a_n| - |a|\big| \leq |a_n - a| < \varepsilon$$

So $\lim_{n\to\infty} |a_n| = |a|$. $\square$

(b.) <u>Disprove</u>: Consider $a_n = (-1)^n$. Then:

- $\{|a_n|\}_{n\geq 1}$ converges: since $|a_n| = 1$ $\forall n \geq 1$.
- $\{a_n\}_{n\geq 1}$ diverges: by lecture $\square$